HDOC DAY 70

PROBLEM

Write a recursive function to calculate the **digital root of a number**. Use a **static** variable to count the number of recursive calls.

CODE

```
#include <stdio.h>


int digitalroot(int n)
{
   static int count = 0;
   count++;


   if (n < 10)
   {
      printf("Recursive calls: %d\n", count - 1);
      return n;
   }


   int sum = 0;


   while (n > 0)
   {
      sum += n % 10;
      n /= 10;
   }
```

```
    return digitalroot(sum);
}

int main()
{
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n < 0)
    {
        printf("Enter a non-negative number.\n");
        return 1;
    }

    printf("Digital Root: %d\n", digitalroot(n));

    return 0;
}
```